Fundamentals of Programming using Python • First Year, FinTech

TP 02 — Practical Work

Where You Write It: Interpreter, Scripts, Notebooks, IDEs

8 exercises • no answers in this sheet

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How to work through this TP

- Do the exercises **in order**. Each one uses only what Chapter 02 and the chapters before it have taught — nothing else is needed, and nothing else is expected.
- **Type the code yourself.** Do not copy and paste. Typing is what builds the memory, and you cannot fix an error you did not write.
- Run your file and compare your output with what the exercise asks for. If they differ, read the error message from the bottom up: the last line names the real mistake.
- Only then open the companion sheet *Solutions — Chapter 02*. Reading the answer before trying teaches you nothing.

❑ **What to hand in** – One Python file per exercise that produces code, named `ex1.py` to `ex8.py`, plus a short written answer for the exercises that ask you to explain rather than to compute.

The 8 exercises

Exercise 1. Three doors (written answer) – Name the three ways of running Python. For each one, give a situation from this course where it is the right choice, and one sentence saying why.

Exercise 2. Interpreter or script (written answer) – Amina opens the interpreter, types twelve lines that compute a loan schedule, gets the right answer, and closes the window. The next day her teacher asks her to run it again with a different rate. What has she lost, and what should she have done instead?

Exercise 3. Your first script (code) – Create a file `ex3.py`. It must hold the fee 1 250 DZD and the VAT rate 0.19 in two named variables, compute the VAT and the total, and print one line reading `Fee 1250 DZD, VAT 237.50 DZD, total 1487.50 DZD`, with both amounts shown to two decimals. Put a comment on the first line saying what the file does.

Hint: you may write a long `print` as two strings on two lines.

Exercise 4. The stale cell (written answer) – In a notebook, a student runs cell 1 (`rate = 0.035`), then cell 2 (`print(250000 * rate)`), which prints `8750.0`. She then edits cell 1 to `rate = 0.04` and runs *only* cell 1. (a) What number is on the screen under cell 2? (b) What number would cell 2 print if it were run now? (c) Why is this dangerous, and what is the cure?

Exercise 5. Which tool for which job (written answer) – For each of these four jobs, say whether you would use a script or a notebook, and give the reason in one sentence. (a) A report that must run automatically every Monday morning. (b) Looking at a new dataset for the first time. (c) Handing in TP 02. (d) Explaining a result to a colleague who wants to see the chart next to the code.

Exercise 6. First-day errors (written answer) – Three students cannot run their first file. Say what has gone wrong in each case and what fixes it. (a) The terminal answers `'python' is not recognised`. (b) The terminal answers `can't open file 'fees.py'`. (c) The file runs but nothing appears, and the editor shows the old output.

Exercise 7. A script you can re-run (code) – Write `ex7.py` for a loan of 400 000 DZD at an annual rate of 6% over 3 years, compound. Put the three inputs in named variables on the first three lines and nothing else; compute and print the final amount to two decimals, and the interest paid to two decimals. A reader must be able to change the rate by editing exactly one character.

Hint: the final amount is $C(1+r)^n$; the interest paid is that minus C.

Exercise 8. Choose an environment (written answer) – Karim has an old laptop, a slow internet connection, and has never programmed. Nadia has a good machine and intends to continue with Python after this course. Yacine cannot install software on the shared machine he uses. Recommend an environment for each, with one sentence of justification, and say what each must watch out for.